Shaker Tarneem Refaat Fawzi

Budapest, Hungary, Open to relocate | +36 70 402 9705 | tarneemshaker@gmail.com
www.linkedin.com/in/tarneem-shaker-33448220a | Portfolio: Tarneem Shaker — Mechanical Engineer

Mechanical Engineering graduate with technical experience in manufacturing, maintenance, and mechanical design. Proficient in SolidWorks, Siemens NX, MATLAB, and Python, with project experience in fatigue analysis, vibration monitoring, and predictive maintenance. Skilled in applying engineering analysis and problem-solving methods to improve equipment reliability and manufacturing processes.

EXPERIENCE

Maintenance and Production Trainee | Teriak Group
June 2025 - September 2025

- Monitored tube-forming, fin-forming, and core-building machines used in automotive heat exchanger manufacturing.
- Performed in-process quality inspections of copper and aluminium tubes and fins, identifying defects and irregularities in component formation.
- Inspected production equipment for abnormal noises, improper fin formation, and unexpected shutdowns, documenting faults and reporting findings to the maintenance department.

EDUCATION

Bachelor of Mechanical Engineering | University of Dunaújváros

June 2026 | GPA: 4.52/5.0

PROJECTS – For more projects and details: https://tarneemshaker.github.io/tarneem-portfolio/projects.html

G-Code Toolpath Generator & Analyzer | Personal Project

- Built a Python command-line tool that generates and analyses CNC G-code for pocket milling operations, applying toolpath logic (multi-pass depth cutting, tool radius compensation) and Python fundamentals (argument parsing, regex-based file parsing, geometry calculations)

Footstep Energy Harvesting System | BSc Thesis Project

- Designed the mechanical assembly of a rack-and-pinion footstep energy harvesting system in SolidWorks, including the loading platform, spring return mechanism, transmission housing, flywheel, and structural frame.
- Performed fatigue life analysis of critical components (springs, gears, shafts, bearings) using S-N curve analysis and Miner's cumulative damage rule applied to a realistic pedestrian load spectrum.
- Applied FMEA to identify critical failure modes and proposed preventive maintenance and design improvements.

Rotating Machinery Fault Detection & Predictive Maintenance | MATLAB

- Developed a MATLAB-based condition monitoring system using vibration analysis and Fast Fourier Transform (FFT).
- Implemented FFT-based detection of rotor imbalance, shaft misalignment, and bearing fault signatures, evaluating classification performance across 24 simulated test cases.
- Created a predictive maintenance model to monitor bearing degradation, estimate remaining useful life (RUL), and generate maintenance recommendations.

TECHNICAL SKILLS

- **CAD & Design:** SolidWorks, Siemens NX, 3D Modeling, Assembly Design
- **Engineering Analysis:** Fatigue Analysis, FMEA, Vibration Analysis, FFT
- **Programming:** MATLAB, Python (Engineering Calculations, Automation)
- **Manufacturing:** CNC/G-Code Fundamentals, Root Cause Analysis, 8D Methodology
- **ERP:** SAP PP / SAP PM – Fundamentals

LANGUAGES

- **Arabic** – Native | **English** – Professional | **French** – Intermediate | **Hungarian** – Beginner